

CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Experienced backend engineer with over 9 years of success designing and deploying scalable web services, APIs, and data pipelines using Python. Demonstrated expertise in building ML-integrated applications, vector database search, and cloud-native infrastructure. Proven track record in developing robust recommendation systems, optimizing backend architectures, and collaborating with cross-functional teams to deliver innovative AI-powered features for global users.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL
- **Frameworks:** FastAPI, Flask, Django, Node.js
- **Databases:** PostgreSQL, MySQL, Redis, Elasticsearch, Qdrant
- **Cloud & DevOps:** AWS, Docker, Kubernetes, Terraform, GitHub Actions
- **AI/ML Integration:** Model serving, embeddings, recommendation systems, vector similarity search
- **Tools & Platforms:** Prometheus, Grafana, CI/CD, Git, JIRA, Agile/Scrum
- **Frontend:** React.js, Next.js, GraphQL (basic integration)
- **Other:** REST APIs, Microservices, Distributed Systems, A/B Testing, TDD

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Designed and deployed scalable REST APIs using Python (FastAPI) to support data-driven functionalities across a B2B platform with global reach.
- Integrated ML models into core backend services for dynamic content ranking and personalized recommendations, improving user engagement KPIs by over 25%.
- Engineered high-performance vector search pipelines using Qdrant and Elasticsearch to enable fast semantic queries over large-scale datasets.
- Developed CI/CD pipelines using GitHub Actions and Docker, ensuring zero-downtime deployments and over 90% code coverage in all services.
- Worked closely with ML engineers to productionize NLP and computer vision models through RESTful interfaces and asynchronous pipelines.
- Migrated legacy Flask applications to FastAPI microservices, resulting in 40% lower latency and improved scalability.

- Deployed containerized services on AWS ECS with Terraform scripts for IAC and ensured robust monitoring using Prometheus and Grafana.
- Led code reviews and mentored junior engineers on API best practices, container security, and backend optimization techniques.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Built and maintained modular APIs in Flask and Django for digital marketing automation platforms, handling over 5M requests daily.
- Integrated Redis and Elasticsearch to support low-latency personalization services and real-time data aggregation for marketing analytics.
- Implemented event-driven systems using RabbitMQ and AWS Lambda to process recommendation data asynchronously.
- Collaborated with data scientists to deploy ML inference endpoints using Flask REST APIs, enabling dynamic targeting and user behavior prediction.
- Improved deployment efficiency by building Docker-based pipelines and introducing static analysis and unit testing in CI workflows.
- Participated in system design sessions to migrate monolithic codebases to scalable microservices with isolated deployment logic.
- Enhanced database performance through query indexing and schema optimization for PostgreSQL-based analytics engines.
- Integrated 3rd-party ML APIs to enrich personalization workflows with demographic prediction and content clustering capabilities.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Developed REST APIs using Django and PostgreSQL to support mobile and web applications focused on user behavior analytics.
- Built dynamic recommendation features using collaborative filtering techniques powered by content similarity models.
- Designed and deployed Redis caching layers to enhance system response times under high load traffic.
- Collaborated with UI developers to expose ML-driven API responses via secure GraphQL endpoints for seamless integration.
- Supported transition to containerized environments by introducing Docker Compose setups and CI/CD automation scripts.
- Maintained and debugged backend services for cross-platform data collection tools deployed across distributed environments.
- Implemented real-time dashboard components by connecting REST endpoints with React components via WebSockets.
- Assisted in writing unit and integration tests to ensure reliable operation and minimize regression during weekly deployments.

Software Developer

Decima Digital

November 2015 – April 2016

Tallinn, Estonia

- Supported Python-based backend development for B2C services involving dynamic search, AI recommendations, and analytics dashboards.
- Integrated Elasticsearch to enable free-text and semantic search on large datasets with relevance tuning and score boosting.
- Deployed APIs on AWS EC2 with basic monitoring scripts to track uptime and latency metrics in staging and production.
- Built RESTful services using Flask for ML model serving and handled data preprocessing in production pipelines.
- Participated in design discussions for backend data architecture supporting future recommendation system expansion.
- Refactored legacy database schemas to support new personalization metadata and embeddings storage.
- Worked closely with data annotation teams to expose endpoint-driven labeling workflows through admin interfaces.
- Helped onboard new team members by maintaining updated backend documentation and deployment guides.

Software Developer Intern

Borna Technology

September 2014 – August 2015

Tallinn, Estonia (On-Site)

- Contributed to internal backend tools using Flask and SQLAlchemy, building dashboard services and data ingestion APIs.
- Wrote Python scripts to automate data import from CSV into normalized PostgreSQL databases for internal reporting.
- Developed REST endpoints to serve statistical summaries and ML-prepared data to internal dashboards.
- Supported deployment automation via shell scripting and Docker for internal test environments.
- Collaborated with QA engineers to build end-to-end test scenarios for backend workflows in test pipelines.
- Designed prototype APIs for experimental content ranking systems powered by early recommendation models.
- Maintained clear documentation for all API endpoints and internal data processing jobs.
- Built proof-of-concept search using TF-IDF and cosine similarity to validate product search logic.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,

2010 – 2014

CERTIFICATIONS

- ❖ AWS Certified Solutions Architect – Associate
- ❖ DeepLearning.AI TensorFlow Developer Specialization
- ❖ MLOps Specialization – Coursera
- ❖ Python for Data Science – edX

PROJECTS

AI-Powered Art Recommendation Platform

- Developed an API-first backend using FastAPI for an ML-integrated recommendation engine that ranked artworks based on user behavior, semantic embeddings, and visual similarity.
- Used Qdrant for vector similarity search and built pipeline for embeddings ingestion from vision transformer models.
- Achieved average API response time of <150ms under peak load using Redis caching and optimized I/O.

ML Service Gateway for Smart Personalization

- Created a service layer to expose multiple ML models (classification, recommendation, segmentation) under unified REST endpoints.
- Enabled seamless switching between versions using feature flags and ensured backward compatibility for consumers.